4	**(a)**		Gravitational field strength at a point is defined as the gravitational force per unit mass acting on a small test mass placed at that point.	A1 A1
	(b)		Gravitational field strength at a point is the negative potential gradient at that point.	A1
	(c)	**(i)**	gravitational field strength $g_P = \frac{GM_E}{r^2} = \frac{(6.67\times10^{-11})(5.98\times10^{24})}{(6.7\times10^{6})^2}$ $= 8.89$ N kg^{-1}	C1 A1
	(c)	**(ii)**	F = mg = 2200 x 8.89 = 19600 N	A1
		(iii)	rate of change of potential energy with distance K_P $= \lvert dU/dr \rvert = F = m\,g = 2200 \times 8.89$ $= 19600$ J m^{-1}	A1
		(iv)	Total KE $= \frac{1}{2}mv^2 = \frac{1}{2}m\left(GM\,r^{-1}\right)$ -- or derive E_T = Total KE + Total GPE $= \frac{GMm}{2r} + \left(-\frac{GMm}{r}\right)$ $= -\frac{GMm}{2r}$ $= -\frac{\left(6.67\times10^{-11}\right)\left(5.98\times10^{24}\right)(2200)}{2(6.7\times10^{6})}$ $= -6.6\times10^{10}$ J	C1 C1 A1